# 7+1: symbols and their definitions

Every symbol used in the notebook 7+1.nb and in its documentation, with its defining equation

Repository: github.com/once-ere/7+1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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**Abstract**

This is a self-contained dictionary. For each symbol it gives the mathematical notation, the Mathematica name in 7+1.nb, the defining equation and its meaning, and, where it exists, the name used in the earlier notebook 4+4.nb.

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## 1. Conventions used in every definition below

- Indices $A, B, K, L \in \{0, 1, \ldots, 7\}$ label the coordinates $x_A$, the octonion units $e_A$ and the Clifford generators; $i, j \in \{1, \ldots, 7\}$; $h \in \{1, 2, 3\}$; $k \in \{4, \ldots, 7\}$ where stated.
- Mathematica lists are 1-based: the entry $\eta_{AB}$ of a metric is `η[[A+1, B+1]]`.
- $1_n$ is the $n \times n$ identity matrix (`IdentityMatrix[n]`, named `ID4`, `ID8`, `ID16`, `ID32`); 0 in a block matrix is a zero block; $M^{\mathsf{T}}$ is the transpose, $\overline{M}$ the complex conjugate, $M^{\dagger}$ the conjugate transpose; $\delta_{AB}$ is the Kronecker delta; $\{X, Y\} = XY + YX$, $[X, Y] = XY - YX$.
- $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is a block matrix (Mathematica `ArrayFlatten[{{a, b}, {c, d}}]`); $M \otimes N$ is the Kronecker product (`KroneckerProduct[M, N]`).
- An index 8 on a family of eight Clifford generators $G_0, \ldots, G_7$ means the ordered product $G_8 = G_0 G_1 \cdots G_7$ (the convention of 4+4.nb).
- $Cl(p, q)$: $p$ generators square to $+1$ and $q$ to $-1$ ($v^2 = Q(v)$, $Q$ of signature $(p, q)$).

## 2. Coordinates and metrics

| symbol | Mathematica | definition | meaning |
|---|---|---|---|
| $X$ | `X` | $X = (x_0, x_1, \ldots, x_7)$ | coordinates; also the components of a symbolic octonion or vector |
| $x_0$ | `x0` | — | 7+1.nb: ordinary time, octonion unit $e_0$. 4+4.nb: hidden space-like dimension |
| $x_1, x_2, x_3$ | `x1, x2, x3` | — | ordinary 3-space (both notebooks); octonion units $e_1, e_2, e_3$ |
| $x_4$ | `x4` | — | 7+1.nb: auxiliary space-like, $e_4 = \ell$. 4+4.nb: ordinary time |
| $x_5, x_6, x_7$ | `x5, x6, x7` | — | 7+1.nb: auxiliary space-like, $e_5, e_6, e_7$. 4+4.nb: auxiliary time-like ("superluminal deflating time") |
| $\eta^{(4,4)}$ | `η4488` | $\mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ | metric of 4+4.nb (space-like +1, time-like −1); split-octonion norm |
| $\eta^{(7,1)}$ | `η71` | $\mathrm{diag}(-1,1,1,1,1,1,1,1)$ | 7+1 metric in the sign convention of 4+4.nb; signature (7,1) |
| $\eta^{(1,7)}$ | `η17` | $-\eta^{(7,1)} = \mathrm{diag}(1,-1,-1,-1,-1,-1,-1,-1)$ | 7+1 metric, octonionic convention: $x^{\mathsf{T}}\eta^{(1,7)}x = \mathrm{Re}(xx)$ |
| $\eta^{AB}_{(1,7)}$ | `η17inv` | $(\eta^{(1,7)})^{-1} = \eta^{(1,7)}$ | inverse metric, raises indices |
| $g^O$ | `gO` | $1_8$ | norm metric of the octonions, $N(x) = x^{\mathsf{T}} g^O x$ |
| $g$ | `gAB44` | $\eta^{(4,4)}$ | norm metric of the split octonions (4+4.nb: `g_AB`) |
| $\eta^{(5,3)}$ | `η53` | $\mathrm{diag}(1,-1,-1,-1,1,1,1,1)$ | $\mathrm{Re}(xx)$ of the split octonions |

## 3. Objects reproduced from 4+4.nb

| symbol | Mathematica | definition | meaning (4+4.nb name) |
|---|---|---|---|
| $\epsilon_{hpq4}$ | `Qa[h,p,q]` | `Signature[{h, p, q, 4}]` | sign of the permutation (Levi-Civita symbol) |
| | `Qb[h,p,q]` | $\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}$ | auxiliary symbol |
| $(s_h)_{pq}$ | `s4by4[h]` | $\epsilon_{hpq4} - (\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4})$ | self-dual antisymmetric 4×4 (`SelfDualAntiSymmetric`) |
| $(t_h)_{pq}$ | `t4by4[h]` | $\epsilon_{hpq4} + (\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4})$ | anti-self-dual antisymmetric 4×4 (`AntiSelfDualAntiSymmetric`) |
| $\sigma$ | `σ` | $\left(\begin{smallmatrix} 0 & 1_4 \\ 1_4 & 0 \end{smallmatrix}\right)$ | split metric on spinor space (`σ`, `σ88`) |
| $\zeta_A$ | `ζ[A]` | $\zeta_0 = 1_8$; $\zeta_h = \left(\begin{smallmatrix} 0 & s_h \\ s_h & 0 \end{smallmatrix}\right)$; $\zeta_{7-h} = \left(\begin{smallmatrix} 0 & t_h \\ -t_h & 0 \end{smallmatrix}\right)$; $\zeta_7 = \zeta_1 \cdots \zeta_6$ | 8×8 building blocks (`ζ[A]`) |
| $\tau_A$ | `τ[A]` | $\tau_A = \zeta_A$ | split-octonion matrices (`τ[A]`) |
| $\bar\tau_A$ | `τbar[A]` | $\bar\tau_0 = 1_8$, $\bar\tau_A = \sigma\tau_A^{\mathsf{T}}\sigma$ | conjugate matrices (symbolized tau-bar) |
| $T_A$ | `T16[A]` | $\left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$, $A = 0, \ldots, 7$ | 16×16 Dirac matrices, $\{T_A, T_B\} = 2\eta^{(4,4)}_{AB} 1_{16}$ (`T16^A[A]`) |
| $T_8$ | `T16[8]` | $T_0 T_1 \cdots T_7 = \mathrm{diag}(-1_8, 1_8)$ | chirality matrix (`T16^A[8]`) |
| $\sigma_{16}$ | `σ16` | $T_0 T_1 T_2 T_3 = \mathrm{diag}(-\sigma, \sigma)$ | (`σ16`) |
| $P_L, P_R$ | `PL44`, `PR44` | $\frac{1}{2}(1_{16} - T_8)$, $\frac{1}{2}(1_{16} + T_8)$ | chiral projectors (`P_L`, `P_R`) |
| $U_h$ | `uBasis[[h]]` | eigenvectors of $\sigma$ divided by $\sqrt{2}$ (order of `Eigensystem[σ]`) | orthonormal eigenbasis of $\sigma$ (4+4.nb: double-struck u) |
| | `evalues`, `evecs` | `Eigensystem[σ]` | eigenvalues $(-1,-1,-1,-1,1,1,1,1)$ and eigenvectors of $\sigma$ |

| symbol | Mathematica | definition | meaning (4+4.nb name) |
|---|---|---|---|
| $u$ | `unit` | $U_8 = (1,0,0,0,1,0,0,0)^{\mathsf T}/\sqrt{2}$, `hUSE = 8` | unit spinor: $\sigma u = u$, $u^{\mathsf T}u = u^{\mathsf T}\sigma u = 1$ (`unit`, `hUSE`) |
| $E_{44}(h)$ | `E44[h]` | $(E_{44})_{Ba} = (U_h^{\mathsf T}\sigma\bar\tau_B)_a$ | vielbein (`E^A_a[h]`) |
| $E_{44}(h)^{-1}$ | `E44inv[h]` | $(U_h^{\mathsf T}\sigma U_h)\,(\eta_{00}^{(4,4)}\tau_0 U_h,\dots,\eta_{77}^{(4,4)}\tau_7 U_h)$ (columns) | inverse vielbein (`E^a_A[h]`) |
| $\varepsilon_B$ | `eA[B]`, `EA` | $B = 1,\dots,8$, `EA = Array[eA, 8]` | split-octonion units; $\varepsilon_B \leftrightarrow \tau_{B-1} \leftrightarrow x_{B-1}$ |
| | `gridx44` | `Grid` of $\varepsilon_B\varepsilon_C = \sum_D (g_{BB}E_{44}\tau_{B-1}E_{44}^{-1})_{DC}\varepsilon_D$ | split-octonion multiplication table (`gridx`) |
| | `gridx44From4p4nb` | the table printed by 4+4.nb, Out[325] | reference for the comparison check |
| $L_A^S$ | `LS[A]` | $g_{AA}\,E_{44}(8)\,\tau_A\,E_{44}(8)^{-1}$ | left multiplication by $\varepsilon_{A+1}$ |
| $x \star_S y$ | `sMul[x, y]` | $\big(\sum_A x_A L_A^S\big)y$ | split-octonion product of coefficient lists |
| $\bar x$, $N_S(x)$ | `sConj`, `sNorm2` | $(x_0,-x_1,\dots,-x_7)$; $x^{\mathsf T}\eta^{(4,4)}x$ | split conjugate and norm |
| | `zeroDivisor` | $(\varepsilon_1+\varepsilon_5)(\varepsilon_1-\varepsilon_5)=0$ | a product of two nonzero split octonions |

# 4. The octonions

| symbol | Mathematica | definition | meaning |
|---|---|---|---|
| $\tau_A^O$ | `τO[A]` | $\tau_0^O = 1_8$; $\tau_A^O = \tau_A$ $(A=1,2,3)$; $\tau_A^O = \tau_A\sigma$ $(A=4,\dots,7)$ | octonion 8×8 matrices |
| $\bar\tau_A^O$ | `τObar[A]` | $g^O(\tau_A^O)^{\mathsf T}g^O = (\tau_A^O)^{\mathsf T}$ | conjugate matrices; $\tau_A^O\bar\tau_B^O + \tau_B^O\bar\tau_A^O = 2g_{AB}^O 1_8$ |
| $E^O$ | `EO` | $(E^O)_{Ba} = (u^{\mathsf T}g^O\bar\tau_B^O)_a$; $E^O = \eta^{(4,4)}E_{44}(8)$ | octonion vielbein (orthogonal) |
| $(E^O)^{-1}$ | `EOinv` | $(E^O)^{\mathsf T} = (u^{\mathsf T}g^O u)(g_{00}^O\tau_0^O u,\dots,g_{77}^O\tau_7^O u)$ | inverse vielbein |
| $e_A$ | `e[A]`, `eList` | $A = 0,\dots,7$, `eList = Array[e, 8, 0]` | octonion units, $e_0 = 1$; $e_A \leftrightarrow \tau_A^O \leftrightarrow x_A$ |
| $L_A^O$ | `LO[A]` | $g_{AA}^O\,E^O\tau_A^O(E^O)^{-1}$ | left multiplication by $e_A$: $e_Ae_K = \sum_L (L_A^O)_{LK}e_L$ |
| | `gridO` | `Grid` of $e_Ae_K$, same formula as `gridx` | octonion multiplication table |
| | `gridOExpected` | table computed by `tools/ref71.py` | reference for the comparison check |
| $xy$ | `oMul[x, y]` | $\big(\sum_A x_A L_A^O\big)y$ | octonion product of coefficient lists |
| $\bar x$ | `oConj[x]` | $(x_0,-x_1,\dots,-x_7)$ | conjugate |
| $N(x)$ | `oNorm2[x]` | $x^{\mathsf T}g^O x = x_0^2+\cdots+x_7^2$ | norm |
| $\mathrm{Re}\,x$ | `oRe[x]` | $x_0$ | real part |
| $x^{-1}$ | `oInv[x]` | $\bar x/N(x)$ | inverse |
| $[x,y,z]$ | `oAssoc[x, y, z]` | $(xy)z - x(yz)$ | associator |
| | `toE[x]`, `ev[A]` | $\sum_A x_Ae_A$; `UnitVector[8, A+1]` | display; coefficient list of $e_A$ |
| | `xs`, `ys`, `zs` | `Array[a, 8, 0]`, `Array[b, 8, 0]`, `Array[c, 8, 0]` | general symbolic octonions |
| | `fanoLines`, `orientedLines` | $\{i,j,k\}$ with $e_ie_j = \pm e_k$; $(a,b,c)$ with $e_ae_b = +e_c$ | the 7 Fano lines |
| | `fanoPicture` | centre $e_1$; vertices $e_2,e_4,e_7$; midpoints $e_6,e_3,e_5$ | picture of the Fano plane |
| | `nonAssociativeTriples` | basis triples with $[e_A,e_B,e_K]\neq 0$ | there are 168 |
| | `xNum`, `yNum`, `zNum` | $e_0+2e_1-e_4+3e_7$, $-2e_0+e_2+e_5$, $e_3+e_6$ | numerical example |
| $\ell$ | `ev[4]` | $\ell = e_4$ | Cayley–Dickson doubling unit, $\ell^2=-1$ |
| $pq$, $\bar p$ | `qMul`, `qConj` | product in $\mathrm{span}\{e_0,\dots,e_3\}$; $(p_0,-p_1,-p_2,-p_3)$ | quaternions ($e_1e_2=-e_3$) |
| $(a,b)$ | `asPair[x]` | $x = a+\ell b$, $a=(x_0,\dots,x_3)$, $b=(x_4,\dots,x_7)$ | octonion as a pair of quaternions |
| | `cdMul[γ]` | $(a,b)(c,d) = (ac+\gamma\,d\bar b,\ \bar ad+cb)$ | Cayley–Dickson; $\gamma=-1$ octonions, $+1$ split |
| | `cdPrompt`, `pm` | $(ac-\bar db,\ \bar ad+cb)$ | the formula of [1A] (fails) |
| | `promptX`, `promptY` | $e_0+e_4=(1,1)$, $e_1=(i,0)$ | its counterexample |
| $s_{AB}$ | `sAB[A, B]` | $-1$ if $A\geq 4$ and $B\geq 4$, else $+1$ | octonion product $= s_{AB}\times$ split product |
| $c$, $\varphi$ | `cc`, `φ` | $c = (1,1,1,1,i,i,i,i)$, $\varphi(x)_A = c_Ax_A$ | isomorphism split$\otimes\mathbb{C}\to\mathbb{O}\otimes\mathbb{C}$ |

# 5. Clifford algebras

| symbol | Mathematica | definition | meaning |
|---|---|---|---|
| $T_A^O$ | `T16O[A]` | $\begin{pmatrix} 0 & \bar\tau_A^O \\ \tau_A^O & 0 \end{pmatrix}$ | $Cl(8,0)$: $\{T_A^O, T_B^O\} = 2\delta_{AB} 1_{16}$ |
| $T_8^O$ | `T16O[8]` | $T_0^O \cdots T_7^O = T_8 = \mathrm{diag}(-1_8, 1_8)$ | volume element of $Cl(8,0)$ |
| $S_{44}$ | `S44` | $T_4 T_5 T_6 T_7 = \mathrm{diag}(\sigma, \sigma)$ | $T_k^O = T_k S_{44}$ for $k \geq 4$ |
| $\Gamma_A^{(1,7)}$ | `Γ17[A]` | $\Gamma_0^{(1,7)} = T_0^O = T_0$; $\Gamma_i^{(1,7)} = T_0^O T_i^O$ $(= T_0 T_i,\ i \leq 3;$ $= T_0 T_i S_{44},\ i \geq 4)$ | REAL 7+1 generators: $\{\Gamma_A^{(1,7)}, \Gamma_B^{(1,7)}\} = 2\eta_{AB}^{(1,7)} 1_{16}$ |
| $\Gamma_8^{(1,7)}$ | `Γ17[8]` | $\Gamma_0^{(1,7)} \cdots \Gamma_7^{(1,7)} = \begin{pmatrix} 0 & -1_8 \\ 1_8 & 0 \end{pmatrix}$ | volume element, square $-1_{16}$ |
| | `products256` | $\Gamma_{A_1}^{(1,7)} \cdots \Gamma_{A_k}^{(1,7)},\ A_1 < \cdots < A_k$ | the 256 basis products |
| $\tau_A^C$, $\bar\tau_A^C$ | `τC[A]`, `τCbar[A]` | $\tau_0^C = \bar\tau_0^C = i1_8$; $\tau_k^C = \tau_k^O$, $\bar\tau_k^C = \bar\tau_k^O$ | complexified 8×8 matrices |
| $\Gamma_A^{(7,1)}$ | `Γ71[A]` | $\begin{pmatrix} 0 & \bar\tau_A^C \\ \tau_A^C & 0 \end{pmatrix}$; $\Gamma_0^{(7,1)} = iT_0$, $\Gamma_k^{(7,1)} = T_k^O$ | COMPLEX generators: $\{\Gamma_A^{(7,1)}, \Gamma_B^{(7,1)}\} = 2\eta_{AB}^{(7,1)} 1_{16}$ |
| $\Gamma_8^{(7,1)}$ | `Γ71[8]` | $\Gamma_0^{(7,1)} \cdots \Gamma_7^{(7,1)} = iT_8$ | volume element, square $-1_{16}$ |
| $S$ | `Ssim` | $(1_{16} - i\Gamma_0^{(1,7)})/\sqrt{2}$ | unitary; $\Gamma_A^{(7,1)} = S(i\Gamma_A^{(1,7)})S^{-1}$ |
| | `intertwiners[gs]` | `NullSpace` basis of $\{M : M\overline{g} = gM\ \forall g\}$ | Schur intertwiners |
| | `indicator[gs]` | sign of the scalar $M\overline{M}$ | $+1$: $\mathbb{R}(16)$; $-1$: $\mathbb{H}(8)$ |
| $M_0$ | `M71` | intertwiners of $\Gamma^{(7,1)}$, spanned by $\Gamma_8^{(1,7)}$ | $M_0\overline{M_0} = -1_{16}$ |
| $\varepsilon$ | `ε2` | $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ | $\varepsilon^2 = -1_2$ |
| $\widehat\Gamma_A^{(7,1)}$ | `Γ71R32[A]` | $\varepsilon \otimes \Gamma_A^{(1,7)}$ | REAL 32×32 generators with $\eta^{(7,1)}$ |
| $Q_i$, $Q_j$, $Q_k$ | `Qi`, `Qj`, `Qk` | $\varepsilon \otimes 1_{16}$, $\sigma_1 \otimes \Gamma_8^{(1,7)}$, $\sigma_3 \otimes \Gamma_8^{(1,7)}$ | quaternion units commuting with all $\widehat\Gamma_A^{(7,1)}$ |
| | `commutantBasis`, `commutantTime` | `NullSpace` of $\{X : X\widehat\Gamma_A^{(7,1)} = \widehat\Gamma_A^{(7,1)} X\}$; its timing | dimension 4: $\mathbb{H}$ |
| $\chi^{(7,1)}$ | `χ71` | $-i\Gamma_8^{(7,1)} = T_8$ | chirality matrix ($\Gamma^{(7,1)}$ basis) |
| | `PL71`, `PR71` | $\frac{1}{2}(1_{16} \mp \chi^{(7,1)}) = P_L, P_R$ | 7+1 chiral projectors |
| $\chi^{(1,7)}$ | `χ17`, `PL17`, `PR17` | $i\Gamma_8^{(1,7)}$; $\frac{1}{2}(1_{16} \mp \chi^{(1,7)})$ | complex chirality and projectors (rank 8) |
| $E_{16}^O$ | `E016` | $\mathrm{diag}(E^O, E^O)$ | rotation to the octonion basis |
| $\Gamma_A^e$ | `Γ17e[A]` | $E_{16}^O \Gamma_A^{(1,7)} (E_{16}^O)^\mathsf{T}$; $\Gamma_0^e = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma_i^e = \mathrm{diag}(L_i^O, -L_i^O)$ | $\Gamma^{(1,7)}$ in the octonion basis |
| $\vec x$, $\psi$ | `xImag`, `ψs` | $(0, x_1, \ldots, x_7)$; `Array[ψ, 8, 0]` | imaginary part; a general octonion |

# 6. Examples and bookkeeping

| symbol | Mathematica | definition | meaning |
|---|---|---|---|
| $\Sigma_{AB}$ | `Σ17[A, B]`, `pairs` | $\frac{1}{4}[\Gamma_A^{(1,7)}, \Gamma_B^{(1,7)}]$; `pairs` lists the 28 pairs $A < B$ | generators of Spin(1,7) |
| $S(\beta)^{\pm 1}$ | `Sβ`, `Sβinv` | $\cosh\frac{\beta}{2}\, 1_{16} \pm \sinh\frac{\beta}{2}\, \Gamma_0^{(1,7)}\Gamma_1^{(1,7)}$ | boost along $x_1$ and its inverse |
| $R(\theta)^{\pm 1}$ | `Rθ`, `Rθinv` | $\cos\frac{\theta}{2}\, 1_{16} \pm \sin\frac{\theta}{2}\, \Gamma_4^{(1,7)}\Gamma_5^{(1,7)}$ | rotation in the $x_4$–$x_5$ plane and its inverse |
| $\Lambda$ | `Λboost`, `Λrot` | $\Lambda_{BK} = \frac{1}{16}\mathrm{tr}(\eta_{BB}^{(1,7)} \Gamma_B^{(1,7)} S \Gamma_K^{(1,7)} S^{-1})$ | Lorentz matrices of the two examples |
| $\Gamma^A$ | `Γ17up[A]` | $\eta_{(1,7)}^{AA} \Gamma_A^{(1,7)}$ | raised index |
| $\Psi$ | `Ψfun` | $(f_1(X), \ldots, f_{16}(X))$, `f[k] @@ X` | 16 arbitrary functions of $x_0, \ldots, x_7$ |
| $\Gamma^A \partial_A$ | `DiracOp17[ψ]` | $\sum_A \Gamma^A\, \partial\psi/\partial x_A$ | 7+1 Dirac operator |
| $\Box$ | `waveOp17[ψ]` | $\sum_A \eta_{(1,7)}^{AA}\, \partial^2\psi/\partial x_A^2$ | 7+1 wave operator |
| $p$, $m$ | `pvec`, `mass` | $p = (3,2,2,0,0,0,0,0)$ (contravariant $p^A$), $m = 1$ | $p^\mathsf{T}\eta^{(1,7)}p = m^2$ |
| $\not{p}$ | `pslash` | $\sum_A p^A \Gamma_A^{(1,7)}$ | $\not{p}^{\,2} = m^2 1_{16}$ |

| symbol | Mathematica | definition | meaning |
|---|---|---|---|
| $W$ | `realAmplitudes` | `NullSpace[pslash - mass ID16]` | eight real spinor amplitudes, $(\not{p} - m)w = 0$ |
| $p \cdot x$ | `pvec.η17.X` | $p^{\mathsf{T}} \eta^{(1,7)} x$ | phase of the plane waves |
| | `Ψwave` | $w\, e^{-i\, p \cdot x}$ | solves $(i\Gamma^A \partial_A - m)\Psi = 0$; complex |
| $J$ | `antiCommutant` | `NullSpace` of $J\Gamma_A^{(1,7)} + \Gamma_A^{(1,7)} J = 0$ $(A = 0, \ldots, 7)$ | spanned by $\Gamma_8^{(1,7)}$ alone, $J^2 < 0$ |
| $k$, $\not{k}$ | `kvec`, `kslash` | $k = (1, 1, 0, 0, 0, 0, 0, 0)$, $\not{k} = \sum_A k^A \Gamma_A^{(1,7)}$ | null momentum, $\not{k}^{\,2} = 0$ |
| | `realNull` | `NullSpace[kslash]` | 8 real solutions of $\not{k}\, w = 0$ |
| | `ΨmasslessReal` | $w \cos(k \cdot x)$ | REAL solution of $\Gamma^A \partial_A \Psi = 0$ |
| $\eta^{AB}_{(7,1)}$ | `η71inv` | $(\eta^{(7,1)})^{-1} = \eta^{(7,1)}$ | raises indices for $\widehat{\Gamma}$ |
| $\widehat{\Gamma}^A \partial_A$ | `DiracOp71R32[ψ]` | $\sum_A \eta^{AA}_{(7,1)} \widehat{\Gamma}^{(7,1)}_A\, \partial\psi/\partial x_A$ | real 32-component Dirac operator |
| | `Ψfun32` | $(g_1(X), \ldots, g_{32}(X))$, `g32[k] @@ X` | 32 arbitrary functions |
| $P$ | `pslash32` | $\sum_A p^A \widehat{\Gamma}^{(7,1)}_A$ | $P^2 = -m^2\, 1_{32}$ |
| $w$ | `w32` | `Array[w, 32]` | 32 arbitrary real constants |
| | `Ψ32` | $w \cos(p \cdot x) + \frac{1}{m} P w \sin(p \cdot x)$ | REAL solution of $(\widehat{\Gamma}^A \partial_A - m)\Psi = 0$ |
| $\Gamma^{(5,3)}_A$ | `Γ53[A]` | $\Gamma^{(5,3)}_0 = T_0$, $\Gamma^{(5,3)}_i = T_0 T_i$ | real $Cl(5,3)$ with $\eta^{(5,3)}$ |
| $\Gamma^{(0,8)}_A$ | `Γ08[A]` | $T^O_8 T^O_A$ | real $Cl(0,8)$ |
| | `gens[p]` | $T^O_A$ for $A < p$, $iT^O_A$ for $A \geq p$ | complex generators of signature $(p, 8-p)$ |
| | `classification`, `claimed1B`, `correct` | $(p, q, \text{indicator})$, $p = 8, \ldots, 0$; claimed and computed classes | classification table |
| | `checks`, `check[name, test]` | association of results; stores `TrueQ[test]` and returns `name -> result` | verification bookkeeping |
| | `failedChecks`, `allChecksPassed` | names of checks that are not True; `AllTrue[Values[checks], TrueQ]` | must be `{}` and `True` |

# 7. Explicit matrices

The 4×4 blocks, the seven split matrices $\tau_A$, and the four new octonion matrices ($\tau^O_A = \tau_A$ for $A \leq 3$):

$$s_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix},\ s_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix},\ s_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix},$$

$$t_1 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},\ t_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},\ t_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

$$\tau_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},\quad \tau_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_3 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_5 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_6 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\tau_7 = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\tau_4^O = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}, \quad \tau_5^O = \begin{pmatrix} 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \end{pmatrix},$$

$$\tau_6^O = \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_7^O = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

# 8. Old names (4+4.nb) and new names (7+1.nb)

| **4+4.nb** | **7+1.nb** | **reason** |
|---|---|---|
| σ, σ88 | σ | unchanged |
| τ[A], ζ[A], s4by4[h], t4by4[h] | unchanged | — |
| symbolized tau-bar | τbar[A] | 2-D symbol made with the Notation package; a plain name needs no package |
| T16^A[A1] (symbolized) | T16[A] | the superscript looked like a power and the letter A was not an index |
| P_L, P_R | PL44, PR44 | the 7+1 projectors get their own names PL71, PR71, PL17, PR17 |
| double-struck u | uBasis | plain name |
| g_AB | gAB44 | subscripted name; plain name |
| E^A_a[h], E^a_A[h] (symbolized) | E44[h], E44inv[h] | the super- and subscripts were part of the name, not indices |
| eA[1..8] (1-based) | e[0..7] (0-based) for the octonions | aligns $e_A$ with $\tau^O_A$ and $x_A$: no $B-1$ shift |
| gridx | gridx44, and the new grid0 | two tables now |